OEA Framework

Full Experiment Insights Report

Structured Recursive Calibration

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1. Pilot Experiment

Recursive stability & epistemic friction (30 seeds, bigram proxy)

Experiment Design

```
Seeds: 30   Iterations: 10   Token space: 12
```

Recursive Stability

```
Control KLD:   2.6603  [2.1234, 3.1971]
OEA KLD:       0.2874  [0.1780, 0.3967]
Control stab:  0.8603  [0.8406, 0.8799]
OEA stab:      0.9809  [0.9779, 0.9839]
Delta:         +0.1206
```

Epistemic Friction

```
Control TRR:   0.6119  [0.6034, 0.6205]
OEA TRR:       0.8440  [0.8384, 0.8495]
Control FRR:   0.2327  [0.2264, 0.2390]
OEA FRR:       0.1210  [0.1165, 0.1256]
TRR delta:     +0.2320
FRR delta:     -0.1117
```

Interpretation

```
H1 supported: OEA stability +12.1% absolute over control.
H2 supported: TRR +23.2pp, FRR -11.2pp.
Scope: bigram proxy harness only.
```

2. Credibility Suite

12-variant ablation study (7,776 runs, d=4.56, p<0.001)

Configuration

```
Total runs: 7,776  (12 variants × 12 seeds × 3 depths × 3 ratios × 3 noise)
Corpora: public_domain_snippets, scientific_snippets
```

Headline: OEA Full vs Control (replace)

```
Stability delta:   +0.4882   (d = 4.56, p < 0.001)
Perplexity delta:  +231.9
TRR delta:         +0.2541
FRR delta:         -0.1600
```

Top 5 Variants by Stability

```
ablation_A                stab=0.9565  TRR=0.635  FRR=0.207
ablation_OE               stab=0.9554  TRR=0.775  FRR=0.120
ablation_O                stab=0.9550  TRR=0.613  FRR=0.222
ablation_EA               stab=0.9496  TRR=0.806  FRR=0.100
ablation_miscalibrated    stab=0.9433  TRR=0.257  FRR=0.651
```

Key Finding

```
oea_full: TRR=0.836, FRR=0.081 (best epistemic discrimination).
ablation_miscalibrated: TRR=0.257, FRR=0.651 (reversed — H2 confirmed).
Stability & epistemic filtering are partially orthogonal.
```

Credibility Suite — Aggregate Metrics

648 runs per variant across 2 domain corpora

Variant	Stability	TRR	FRR	Perplexity	N
ablation_A	0.9565	0.635	0.207	725.0	648
ablation_E	0.9347	0.752	0.134	452.7	648
ablation_EA	0.9496	0.806	0.100	436.4	648
ablation_O	0.9550	0.613	0.222	986.1	648
ablation_OA	0.9378	0.667	0.188	654.5	648
ablation_OE	0.9554	0.775	0.120	531.7	648
ablation_miscalibrated	0.9433	0.257	0.651	891.7	648
baseline_calibration_only	0.9347	0.651	0.198	452.7	648
baseline_rag_only	0.9347	0.682	0.178	452.7	648
control_accumulate	0.9433	0.582	0.241	891.7	648
control_replace	0.4485	0.582	0.241	270.2	648
oea_full	0.9367	0.836	0.081	502.2	648

3. Real LLM Validation

Three models, two architecture families (GPT-2, GPT-Neo)

Design

```
3 models × 4 variants × 10 seeds × 10 iterations = 1,200 steps
Hardware: RTX 4070 SUPER, CUDA 12.1
```

distilgpt2 (82M)

```
Control (iter 10): log_prob=-0.911  JSD=0.459  ROUGE-L=0.046
oea_anchored               log_prob=+0.6160  JSD=-0.0118  ROUGE-L=-0.0085
oea_miscalibrated          log_prob=-1.1439  JSD=+0.1027  ROUGE-L=0.0081
oea_rag_only               log_prob=-0.5353  JSD=+0.0769  ROUGE-L=-0.0009
CQ measurement:
  oea_rag_only: TRR=0.5625 → CQ=0.501
  oea_anchored: TRR=0.5787 → CQ=0.520
  oea_miscalibrated: TRR=0.6238 → CQ=0.571
```

gpt2 (124M)

```
Control (iter 10): log_prob=-1.916  JSD=0.363  ROUGE-L=0.0474
oea_anchored               log_prob=+1.6255  JSD=-0.0742  ROUGE-L=-0.016
oea_miscalibrated          log_prob=-0.5539  JSD=+0.0207  ROUGE-L=0.0071
oea_rag_only               log_prob=-0.0755  JSD=+0.0006  ROUGE-L=0.0016
CQ measurement:
  oea_rag_only: TRR=0.6312 → CQ=0.614
  oea_anchored: TRR=0.4700 → CQ=0.445
  oea_miscalibrated: TRR=0.6250 → CQ=0.607
```

gpt-neo-125M (125M)

```
Control (iter 10): log_prob=-1.055  JSD=0.408  ROUGE-L=0.0489
oea_anchored               log_prob=+0.8163  JSD=-0.0341  ROUGE-L=-0.0122
oea_miscalibrated          log_prob=-1.3664  JSD=+0.0610  ROUGE-L=0.0047
oea_rag_only               log_prob=-0.4288  JSD=+0.0098  ROUGE-L=-0.0023
CQ measurement:
  oea_rag_only: TRR=0.5475 → CQ=0.493
  oea_anchored: TRR=0.4987 → CQ=0.438
  oea_miscalibrated: TRR=0.5725 → CQ=0.521
```

Hypothesis Summary (all 3 models)

```
H1 (Calibration Direction): SUPPORTED  (+0.62 to +1.63 nats)
H2 (Miscalibration Reversal): SUPPORTED  (-0.55 to -1.37 nats)
H3 (RAG without Filter): SUPPORTED  (-0.08 to -0.54 nats)
ROUGE-L dissociation: CONFIRMED across all 3 models
```

Real LLM — Final Iteration Results (iter 10)

10 seeds × 10 iterations per model/variant

Model	Variant	Log-prob	Δ Log-prob	JSD	ROUGE-L
distilgpt2	control	-0.911	—	0.459	0.0460
	oea_anchored	-0.295	+0.6160	0.471	0.0375
	oea_miscalibrated	-2.055	-1.1439	0.356	0.0542
	oea_rag_only	-1.447	-0.5353	0.382	0.0450
gpt2	control	-1.916	—	0.363	0.0474
	oea_anchored	-0.290	+1.6255	0.438	0.0313
	oea_miscalibrated	-2.470	-0.5539	0.343	0.0545
	oea_rag_only	-1.991	-0.0755	0.363	0.0490
gpt-neo-125M	control	-1.055	—	0.408	0.0489
	oea_anchored	-0.239	+0.8163	0.442	0.0367
	oea_miscalibrated	-2.421	-1.3664	0.347	0.0536
	oea_rag_only	-1.484	-0.4288	0.398	0.0466

4. Agentic Memory Drift Benchmark

30-step recursive summarization (20 seeds, bigram proxy)

Design

```
20 seeds × 2 variants × 30 steps
Bigram proxy recursive summarization
```

Final Step (step 30) Comparison

```
uncontrolled:
  entity_retention          0.1571  [0.1550, 0.1592]
  semantic_drift            0.4937  [0.4823, 0.5051]
  hallucination_proxy       0.0000  [0.0000, 0.0000]
  vocab_collapse            0.9213  [0.9089, 0.9336]

oea_controlled:
  entity_retention          0.1561  [0.1535, 0.1587]
  semantic_drift            0.4873  [0.4787, 0.4958]
  hallucination_proxy       0.0000  [0.0000, 0.0000]
  vocab_collapse            0.9150  [0.8999, 0.9301]
```

OEA vs Uncontrolled Deltas

```
  entity_retention          -0.0011 ↓
  semantic_drift            -0.0064 ↓
  hallucination_proxy       +0.0000 ↓
  vocab_collapse            -0.0063 ↓
```

Interpretation

```
Bigram harness provides structural support for H4 only.
Effect sizes |Δ| < 0.01 reflect vocabulary constraints.
Hallucination proxy = 0.0 for both (bigram limitation).
Neural validation required for quantitative conclusions.
```

5. Baseline Competition

OEA vs 5 non-OEA controls (20 seeds, bigram proxy)

Design

20 seeds × 10 iterations

OEA vs 5 non-OEA controls: temperature_low, top-k, entropy_filter, repetition_penalty, rag_only.

Final Iteration Results

Variant	Stability	TRR	FRR	d(stab)	p(stab)
oea_full	0.1550	0.746	0.145	+0.000	1.000
temperature_low	0.1601	0.605	0.226	-0.135	0.668
top_k	0.1656	0.605	0.226	-0.312	0.307
entropy_filter	0.1550	0.630	0.211	+0.000	1.000
repetition_penalty	0.1597	0.605	0.226	-0.127	0.699
rag_only	0.1425	0.590	0.234	+0.317	0.328

Key Finding

OEA achieves highest TRR (0.746 vs next-best entropy_filter 0.630).

Stability differences not significant at N=20 (p > 0.3).

OEA is not reducible to generic constraint tightening.

6. Error Taxonomy

Mean error counts per variant (648 runs each)

Variant	Tail Loss	Novel Noise	False Accept	False Reject
ablation_A	0.0	337.6	109.5	62.1
ablation_E	0.0	0.0	74.3	40.1
ablation_EA	0.0	0.0	58.3	30.0
ablation_O	0.0	522.0	116.2	66.5
ablation_OA	0.0	164.2	100.0	56.3
ablation_OE	0.0	0.0	67.5	35.9
ablation_miscalibrated	0.0	522.0	223.0	195.2
baseline_calibration_only	0.0	0.0	104.7	59.3
baseline_rag_only	0.0	0.0	95.4	53.5
control_accumulate	0.0	522.0	125.5	72.3
control_replace	389.5	239.4	125.5	72.3
oea_full	0.0	0.0	49.2	24.3

CQ Measurement & Evidence Chain

Calibration quality measured from real LLM true-reject rates

CQ Measurement Summary (REQ-OEA-012)

Model	Variant	TRR	CQ
distilgpt2	oea_rag_only	0.5625	0.501
distilgpt2	oea_anchored	0.5787	0.520
distilgpt2	oea_miscalibrated	0.6238	0.571
gpt2	oea_rag_only	0.6312	0.614
gpt2	oea_anchored	0.4700	0.445
gpt2	oea_miscalibrated	0.6250	0.607
gpt-neo-125M	oea_rag_only	0.5475	0.493
gpt-neo-125M	oea_anchored	0.4987	0.438
gpt-neo-125M	oea_miscalibrated	0.5725	0.521

Evidence Chain Status

Design estimate CQ = 0.83 (credibility_suite.py).
Measured CQ for oea_anchored: 0.438–0.446 across models.
Mismatch documented as UNK-001. Direct ECE validation is future work.

Interpretation

Dynamic threshold adapts proportionally to anchoring,
so TRR does not improve despite better log-probability.
This is a known limitation, not a falsification of OEA.